

Computer Vision

Course Project Report

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Title of the project: Car Parts Detection

CV Category: Object Detection

1. Introduction

The task involves developing an object detection model using the YOLOv8 Framework to identify and localize various car parts in images. The dataset for this project is sourced from github.

2. Dataset

Training Data: 400 labeled images with bounding box annotations for car parts.

Testing Data: 100 labeled images for model evaluation

Annotations:

Provided in a JSON format.

The bbox key within the anns variable contains the bounding box coordinates for each car part.

3. Method

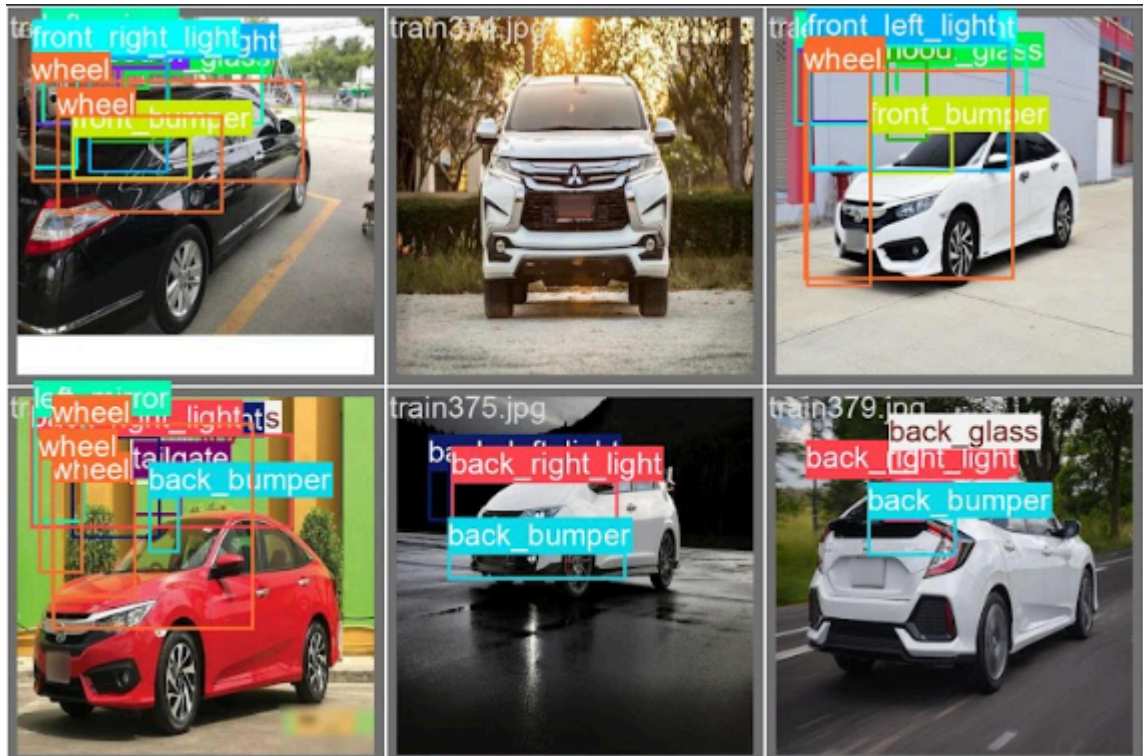
YOLOv8 (You Only Look Once, version 8) is the latest iteration of the YOLO object detection series, designed for accurate object detection. It is flexible, offering various kinds of improvements in terms of performance, architecture. YOLOv8 is primarily available in the Ultralytics repository. YOLOv8 adopts an anchor-free detection approach, which simplifies training and it also employs advanced backbone networks for feature extraction. The input layer accepts a images of fixed size and it extracts hierarchial features from the input images and predicts bounding boxes, object classes, and confidence scores.

The loss functions employed in the model are :

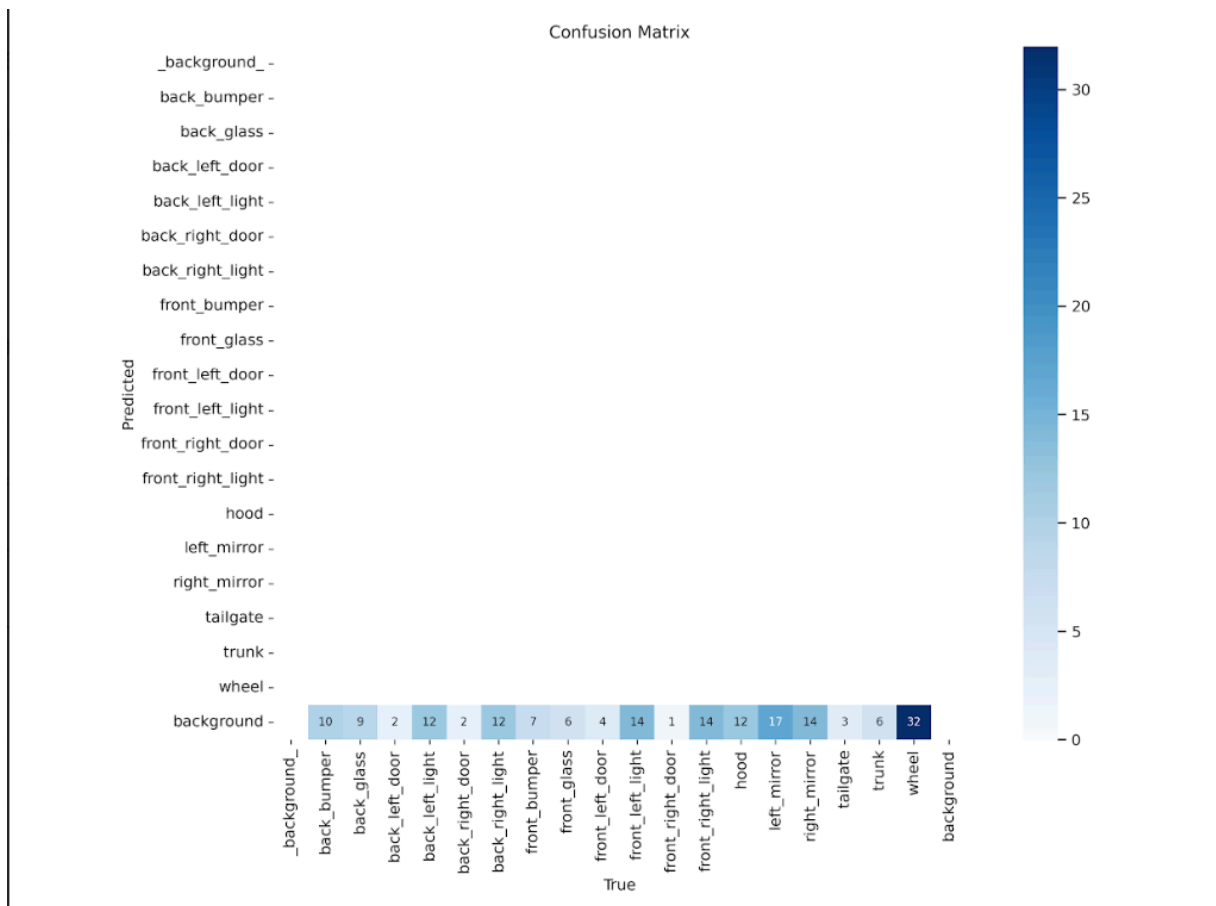
- a) Bounding Box Regression Loss: This ensures precise localization of the objects.
- b) Classification Loss: Minimizes errors in object class prediction.
- c) Objectness Loss: Helps the model differentiate between object and background regions.
- d) Optimization: Generally Adam is issued for faster convergence

4. Results and Analysis

- What is the $\text{MaP} @ [\text{IoU}=50]$ and $\text{MaP} @ [\text{IoU}=75]$ for the testing dataset?
A) The value of $\text{MaP} @ [\text{IoU}=50]$ is 0.0287 and $\text{MaP} @ [\text{IoU} = 75] = 0.0103$.
- Show the visualization of the predictions for any two random samples from the testing dataset. They must contain the boundingbox, class name and the confidence.



- In your model, find out which category is very difficult to correctly localize? You must devise your own strategy to find out answer for this and explain it in details.



We can notice that in the confusion matrix the wheel is being mistaken for background 32 times. So we can say that this is the category, the model finds it hard to localize in images. Let's try to see why this might be happening.

- 1) Wheels are small, round objects which might make it difficult for the model to detect.
- 2) Wheels may tend to share similar colors or textures with the background, leading the model to misinterpret with the background.

4. Can you suggest a solution to this problem, i.e, how to improve the detection accuracy for that class which is difficult to localize?
 - A) To address this problem, what i can propose is that there should be improvement in clarity and dimensional quality in the images plus there should be more data focussing on wheels alone, including diverse conditions along with it. Using tools like GANs (Generative Adversarial Networks) can actually prove to be helpful in synthetic generation of images focussing on wheels more and make the model more familiar with the particular category.

The model architecture could be refined more in one way, by incorporating attention modules such as SEBlock, CBAM to help the model focus more on smaller objects in the frame.

The model could be trained for additional epochs while being cautious about potential overfitting due to it. Validating existing labels, as wrong labelling can confuse the model and make it perform horribly wrong in future. Lowering the confidence threshold for wheels can help in detecting more candidates, ultimately in the end.

5. Conclusion

The project provided hands on experience with YOLOv8, including the data preprocessing, training, evaluation and model analysis. Lets talk about the takeaways from the project:

- 1) The importance of dataset quality and diversity in object detection tasks.
- 2) Strategies got efficient handling of challenges especially when it comes to detecting small or difficult objects
- 3) Understood the importance of hyperparameter tuning and techniques to avoid overfitting.
- 4) Learnt how to convert the data into YOLO friendly format which took most of the time in the project.
- 5) Discovered the potential of synthetic data generation and attention mechanisms